

SABARAGAMUWA UNIVERSITY OF SRILANKA
FACULTY OF COMPUTING
DEPARTMENT OF COMPUTING AND INFORMATION SYSTEMS
IS4103 – OS(P)
ASSIGNMENT 02
DURATION- 30 MINUTES

Scenario:01

An enterprise application runs asynchronous logging operations. To test how the system manages simultaneous background tasks without blocking the user interface, you need to simulate two processes running concurrently with different execution timelines.

Requirements

Write a C program that forks the application into a parent and a child execution thread:

1. The Child process must instantly print its own Process ID and its parent's Process ID using the format: [Child] My PID is X, and my Parent's PID is Y.
2. The child process must then simulate a slow data-transmission delay by suspending its execution for exactly 4 seconds. After waking up, it must print [Child] Data transmission finished. Exiting.
3. The Parent process must immediately print its own Process ID using the format: [Parent] Monitoring Active. PID: X.
4. The parent must not run ahead; it must explicitly block and pause execution until the child process finishes its execution to prevent creating a Zombie process.
5. Once the parent detects the child has exited, it should print [Parent] Child process reaped successfully. Application exiting.